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Medical Receipt OCR Benchmark

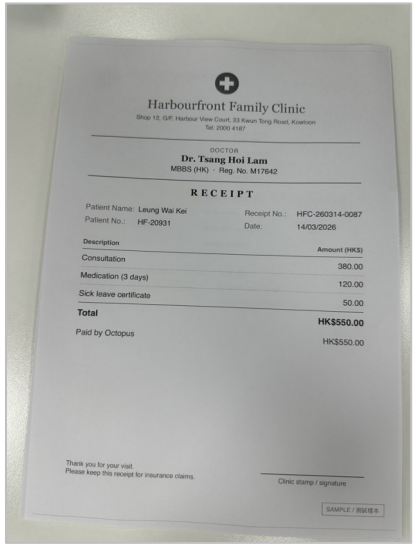
Can Azure OpenAI replace hand-written extraction rules?

9 receipts · 6 fields · manual rules vs AI extraction

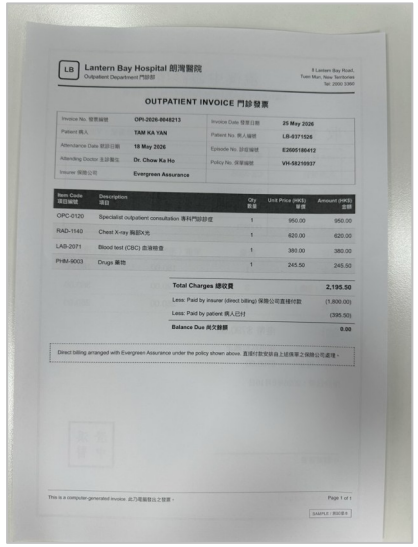
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What We Tested

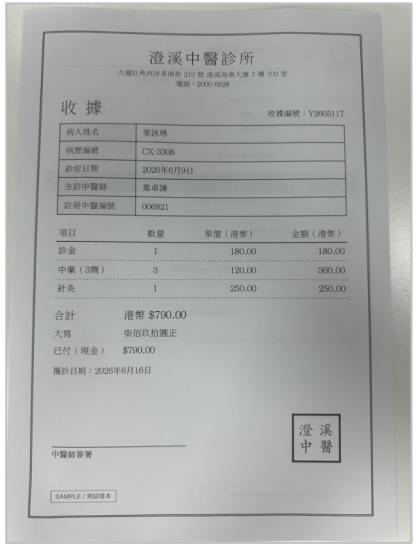
9 receipts with different layouts and languages. Each approach reads the text, then finds 6 fields.



English



Bilingual



Chinese

Approach	1. Read text	2. Find fields
PaddleOCR Light	Open-source OCR	Rules
Azure DI	Azure OCR	Rules
Azure DI + AOAI	Azure OCR	Azure OpenAI

Fields: doctor, registration no., patient, receipt no., service date, total.
51 fields scored.

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Initial Results

Before any tuning on these 9 receipts, AI extraction was far ahead of the first rule sets.

Approach	Accuracy	Time / receipt
PaddleOCR Light + rules	**37.3%** (19/51)	13.9 s
Azure DI + rules	**41.2%** (21/51)	3.8 s
Azure DI + AOAI	**88.2%** (45/51)	6.2 s

Accuracy = correct fields out of 51. The AOAI prompt was written before these receipts. All approaches processed all 9 receipts.

Why the rules scored low

- The original rules knew English labels only: no Chinese or bilingual labels.
- On Chinese and bilingual receipts the OCR read the values, but the rules could not find them.
- **So this compares initial implementations, not the best each approach can do.**

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How We Improved Accuracy

Both approaches improved after reviewing these receipts. Changing rules took far more work than changing a prompt.

Approach	What we changed	Development score	Fixed / broke
PaddleOCR Light + rules	Rewrote the rules by hand: Chinese and bilingual labels, distractor labels, layouts (69 → 225 label phrases)	**88.2%** (45/51)	**+27** / **−1**
Azure DI + rules	The same rule changes, on Azure DI text	**86.3%** (44/51)	**+24** / **−1**
Azure DI + AOAI, prompt v2	Revised the prompt: clearer field definitions, text kept in page rows, a few general checks	**96.1%** (49/51)	**+4** / −0

Changing rules

Code and vocabulary for every new wording or language, then a retest of every receipt. One change broke a receipt that was already correct.

Changing the AI prompt

Plain-language edits to one prompt, then a rerun. No field broke on these receipts.

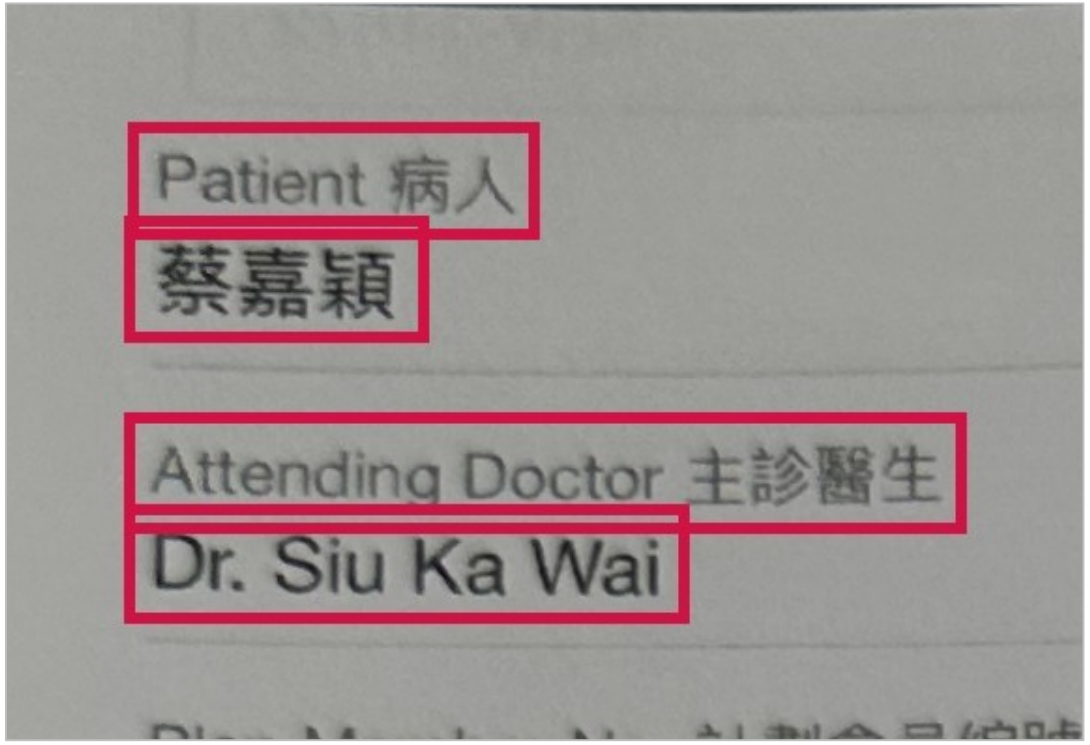
Development results: every change was made after reviewing these 9 receipts, so these scores are not independent validation.

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Example 1: The OCR Was Right, the Rules Were Wrong

Both OCR engines read the names. The original rules took the label 病人 as the patient's name.



Field	Correct	Original rules	Improved rules	DI + AOAI
Patient	蔡嘉穎	✗ 病人	✓ 蔡嘉穎	✓ 蔡嘉穎
Doctor	Dr. Siu Ka Wai	✗ not found	✓ Dr. Siu Ka Wai	✓ Dr. Siu Ka Wai

- Original rules: took the Chinese label 病人 ("patient") as the patient's name.
- Improved rules: fixed by adding bilingual labels by hand.
- AOAI: correct with its original prompt; no change needed.

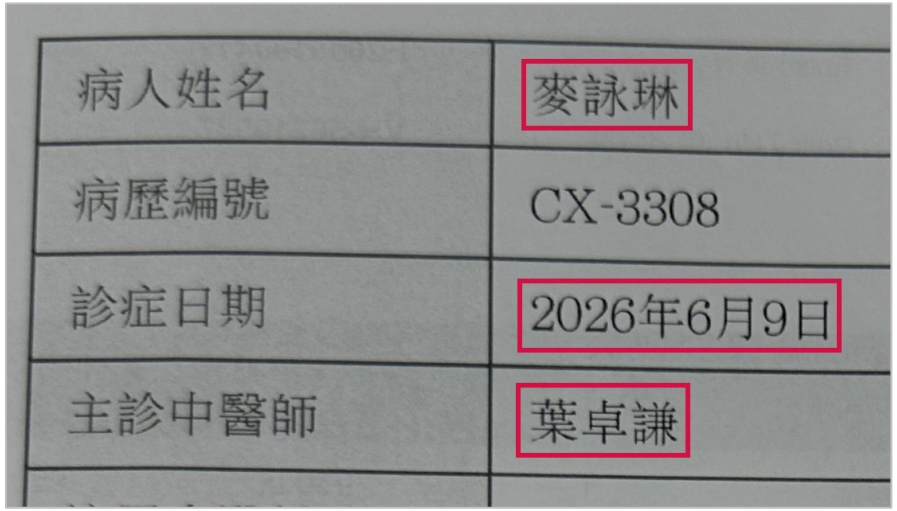
Bilingual medical centre receipt. Rules shown on PaddleOCR Light text; Azure DI with the original rules made the same patient error. DI + AOAI = original prompt.

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Example 2: Every New Layout Meant Manual Rule Work

Each label wording was added by hand, and one change broke a receipt that was already correct.



Chinese medicine receipt: original rules 0/6 fields; after Chinese labels were added by hand 6/6; DI + AOAI 6/6 (original prompt).

69 → 225

label phrases in the rules, added by hand after reviewing the errors

369303 → 380048

vet invoice receipt number: correct before the changes, wrong after (a regression)

New layouts

will likely need more rule changes and a retest of every receipt

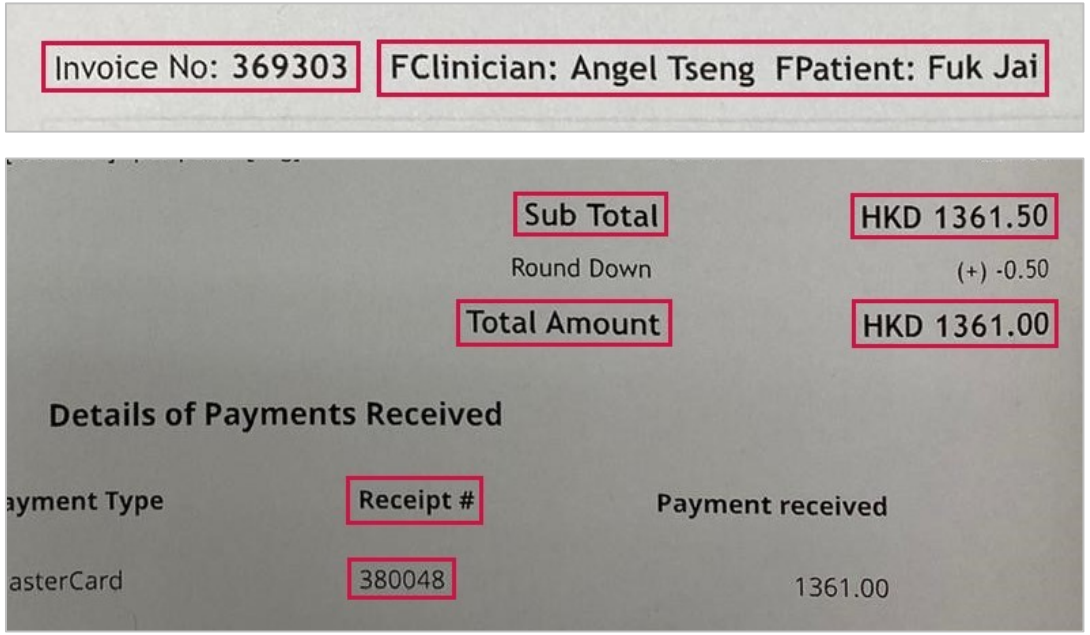
With its original prompt, DI + AOAI needed none of these changes for these receipts. Whether that holds for new layouts still needs to be tested.

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Example 3: AOAI Still Makes Mistakes

On the vet invoice, the improved rules and AOAI each got a different field wrong.



Field	Correct	Original rules	Improved rules	DI + AOAI
Receipt no.	369303	✓ 369303	✗ 380048	✓ 369303
Patient	Fuk Jai	✗ not found	✗ not found	✓ Fuk Jai
Total	1361	✓ 1361.0	✓ 1361.0	✗ 1361.5

- Improved rules: picked the payment receipt number.
- Rules: the OCR read "FPatient:", so the patient label was missed.
- AOAI (original prompt): picked the sub total. Prompt v2 fixes it, but was tuned on this receipt.

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Development Effort and Maintenance

The difference is the manual work to handle each layout, and to keep it working.

	PaddleOCR Light + rules	Azure DI + AOAI
Effort to improve	Rules rewritten by hand after reviewing errors on these receipts	One prompt revision after reviewing errors
What a change looks like	Code and label lists (69 → 225 phrases)	Prompt wording and a few general checks
Side effects seen	1 correct field broken by the rule changes	0 correct fields broken by the prompt revision
New receipt layouts	Likely more rule updates and a regression test of every receipt	Possibly no change; not yet tested
Time and running cost	13.9 s, own compute, no API fee	6.2 s, Azure fees per receipt (to confirm)

The main question is not API price but who maintains the extraction logic as layouts change. Lower maintenance for AOAI is implied by this benchmark, not yet proven.

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Manual Rules vs. AI Extraction: Pros and Cons

"Measured" = this benchmark; the rest is expected.

Manual extraction rules (PaddleOCR Light)

New formats need manual rule updates, which can break working cases and add maintenance and testing.

Pros

- Full control; transparent and predictable on formats the rules know.
- No API fees: open-source OCR on our own compute.
- **Measured:** the OCR read the values; the misses were missing labels.

Cons

- Each new format or language needs rule updates by hand.
- **Measured:** a rule change broke a receipt that was already correct.
- **Measured:** 69 → 225 label phrases added by hand.

AI extraction (Azure DI + AOAI)

Less rule maintenance expected: accuracy depends mainly on the AI model, OCR output and prompt.

Pros

- **Measured:** highest accuracy before any tuning on these receipts.
- Reads varied formats using existing model capabilities.
- **Measured:** faster, 6.2 s vs 13.9 s per receipt.

Cons

- **Measured:** still makes mistakes, e.g. a sub total taken as the total.
- Less direct control: results depend on model, OCR, prompt.
- API fees; needs validation, monitoring and prompt upkeep.

Neither is maintenance free. The difference is where the effort sits: rules for every layout, or one prompt plus validation.

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Conclusion and Next Steps

The real trade-off: maintain extraction rules ourselves, or let AI replace that layer.

- Initial implementations: AI extraction led by a wide margin, mainly because the first rules had no Chinese support.
- After reviewing these receipts both improved. Rules needed code and vocabulary changes; AI needed one prompt revision. Neither is validated on unseen receipts.
- **So the choice is whether to keep writing and maintaining extraction rules, or to use AI for that layer.**

1. Freeze both

Lock the improved rules and AOAI prompt v2.

2. Test on unseen receipts

30+ real, anonymised receipts. Measure accuracy and count the rule changes needed.

3. Then compare cost

Compute, Document Intelligence pages and gpt-5-mini tokens.

Limitation: 9 receipts, mostly printed mock receipts; the rules and prompt v2 were tuned on them. AOAI still makes mistakes and has not been tested on a large unseen set.

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